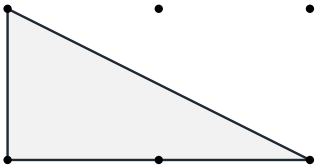
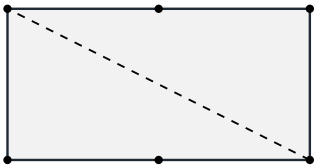
Join dots with straight sides to make one closed outline. Sides cannot cross or touch except at neighboring corners. The area of one grid square is 1. Find or check areas with pieces, cuts or copies.

Input



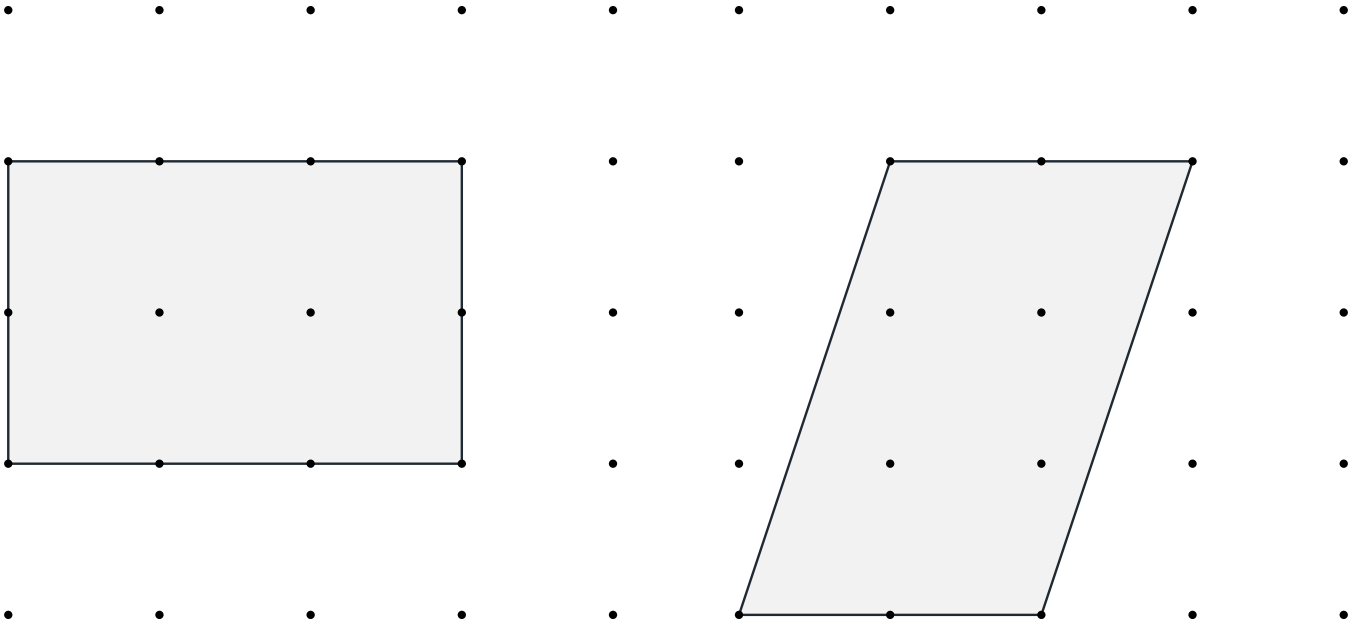
Two copies joined



Output

Rectangle area: 2
One copy's area: 1

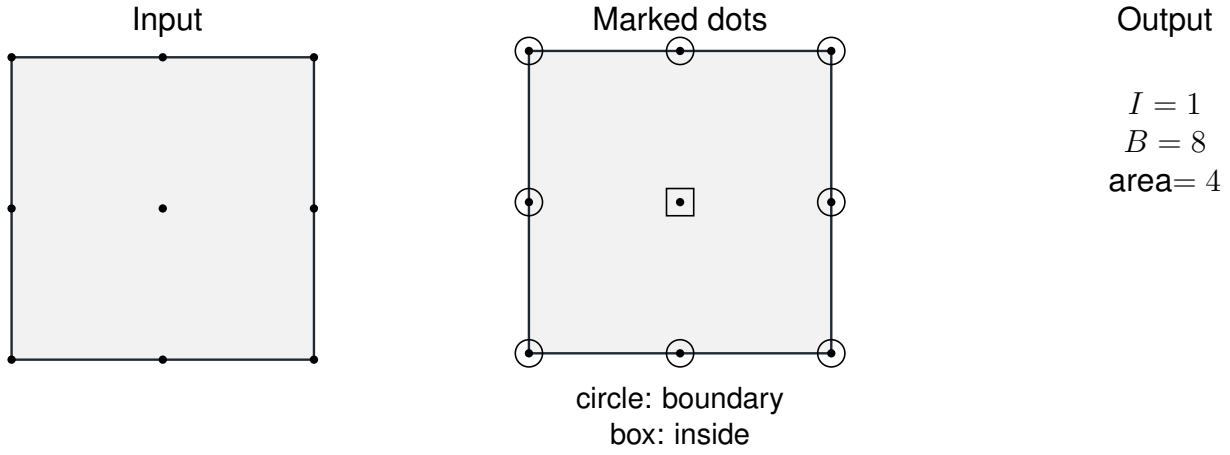
Problem 1: Find the exact area of each shape.



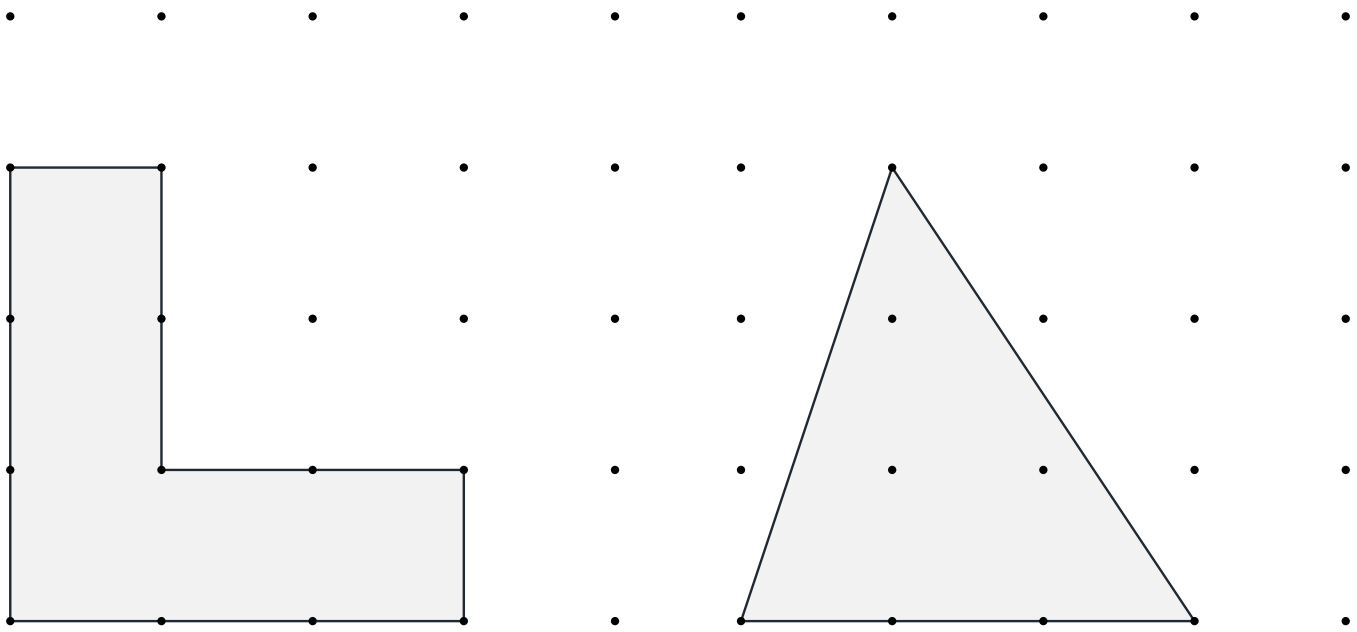
area=_____

area=_____

I counts dots strictly inside. B counts every dot on the outline, including dots between corners. Count each dot once.



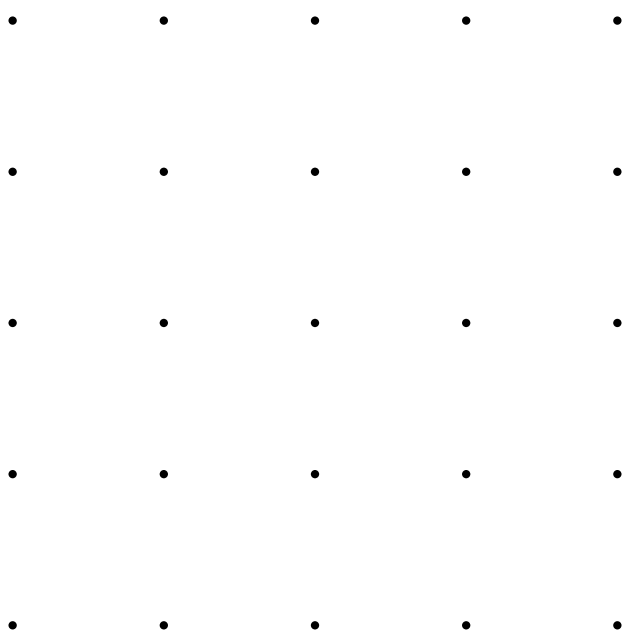
Problem 2: For each shape, mark the dots you count and make an $I, B,$ area record.



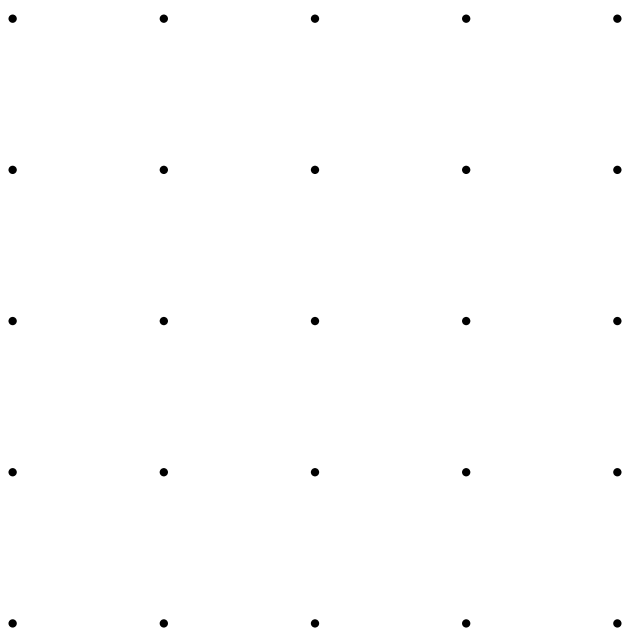
$I =$_____ $B =$_____ area=_____

$I =$_____ $B =$_____ area=_____

Problem 3: Make a triangle and a four-sided shape, each with $I = 2$ and $B = 6$. Find their areas.



$I =$_____ $B =$_____ area=_____



$I =$_____ $B =$_____ area=_____

Problem 4: Find a rule that predicts area from I and B. Use all your records. Draw a new shape that could defeat your rule.



$I =$_____ $B =$_____ area=_____

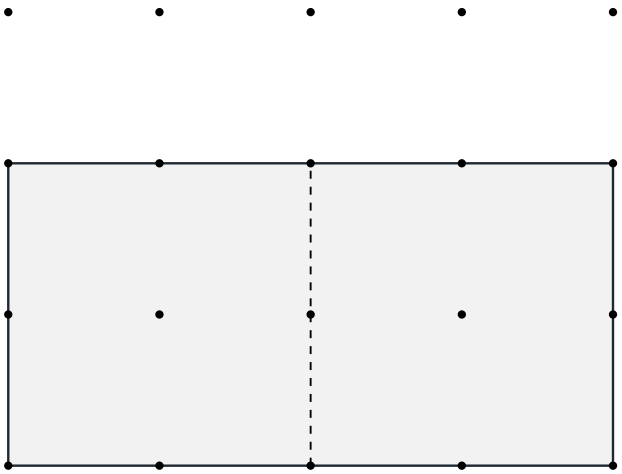
Problem 5: Pick's rule for the allowed polygons is

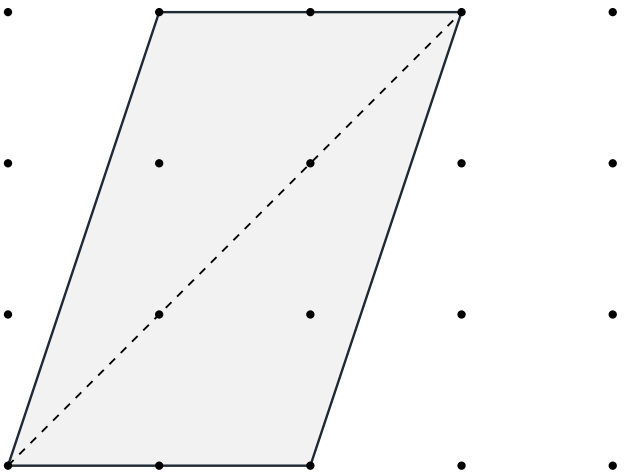
$$\text{area} = I + (\text{half of } B) - 1.$$

Make shapes of area 6 with as many different (I, B) records as you can.



Problem 6: Each dashed seam cuts a shape into two pieces. Make an $I, B,$ area record for each piece and for the whole shape. What changes when the seam disappears?





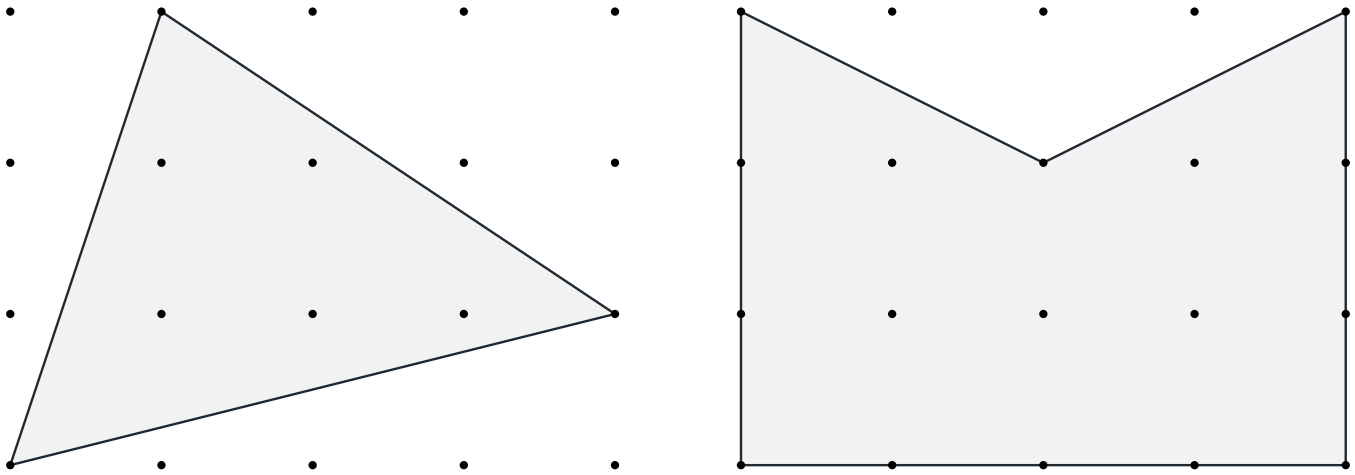
Left piece:
$I =$_____ $B =$_____ area=_____
Right piece:
$I =$_____ $B =$_____ area=_____
Whole:
$I =$_____ $B =$_____ area=_____

Lower piece:
$I =$_____ $B =$_____ area=_____
Upper piece:
$I =$_____ $B =$_____ area=_____
Whole:
$I =$_____ $B =$_____ area=_____

Problem 7: Join two allowed polygons along one whole side. They must meet only on that side, and the joined shape must obey the shared rules. Explain why the two values $I + (\text{half of } B) - 1$ add to the value for the whole. Your explanation must work however many dots lie on the shared side.

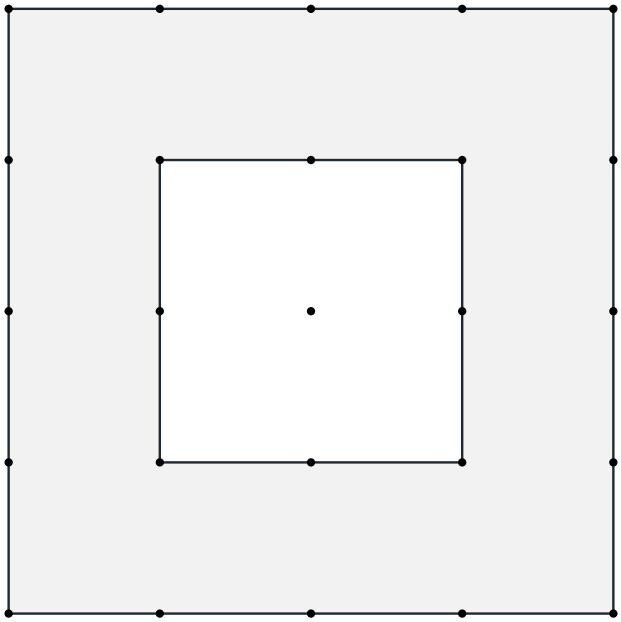


Problem 8: Explain why Pick's rule works for every allowed polygon, including these two.

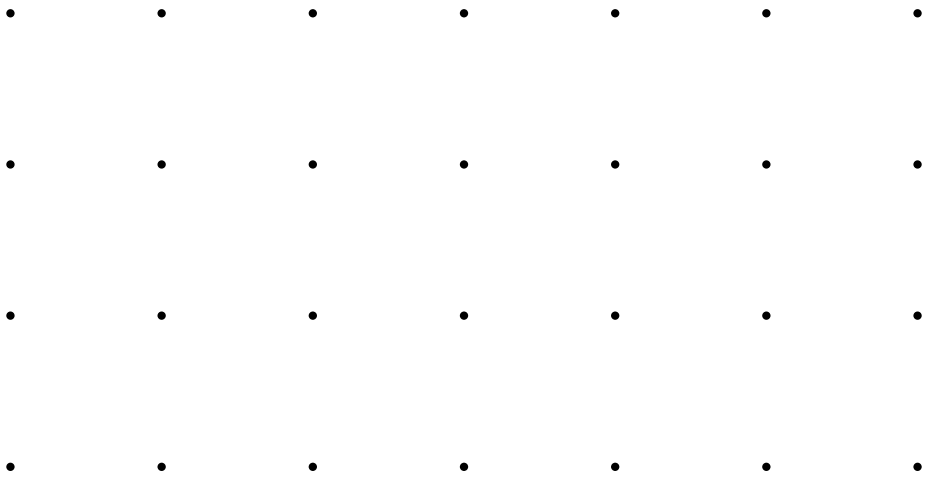


Keep each hole inside the outer outline. All boundary loops must stay separate, with straight sides and corners on dots. Count dots on all outlines in B; dots in a hole do not count in I.

Problem 9: The shaded shape has a square hole. Find its area, I and B. How must Pick's rule change when there is one hole? Test your change on another shape with one hole.



$I =$_____ $B =$_____ area=_____



Problem 10: Make a connected shape with two holes. Find area without using dot counts. Then find a dot-count rule for a connected shape with any number of holes.



$I =$_____ $B =$_____ area=_____